

Urban Dynamics Unveiled: Prompt-driven Building Fine-grained Change Detection

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Abstract—Building fine-grained change detection (BFCD) aims to identify the concrete change statuses of changed buildings. Despite its importance, BFCD has received limited attention, with few specialized models and benchmark datasets available. In this work, we construct a manually annotated BFCD dataset, termed UEDC-BFCD, to advance research in this direction. It introduces three statuses of changed buildings: disappeared, newly appeared, and replaced regions. Meanwhile, it encompasses multiple event types, including Urban Expansion, natural Disasters, and regional Conflicts. In addition, a new Prompt-driven Change Detector (PromptCD) is developed, tailored for BFCD. First, we introduce the concept of Single-Temporal Change Embedding (STCE), enabling the differentiation of changed buildings from unchanged regions in the single-temporal image. PromptCD combines bi-temporal absolute difference representations with single-temporal features to form STCE. Then, we develop a Prompt-Guided Transformer Decoder (PGTD). It employs two learnable prompts to interact with the multi-level STCEs during decoding, thus refining the STCEs and modeling the spatiotemporal correlation. In particular, PGTD is computationally economical, ensuring the efficiency of the model. Finally, the change statuses of buildings are determined by combining the changed regions of buildings in the pre-temporal and post-temporal images. Extensive experiments conducted on the UEDC-BFCD dataset, its urban expansion-focused subset UE-BFCD, as well as two existing datasets, demonstrate that PromptCD not only achieves superior performance but also maintains competitive complexity.

Index Terms—Building change detection, prompt-guided decoding, fine-grained, change status.

I. INTRODUCTION

Building change detection (BCD) is a crucial research area in computer vision and remote sensing, aimed at identifying changes in buildings by analyzing bi-temporal images captured by satellites or aircraft over the same location at different time points [1]–[3]. Deep learning techniques have been extensively explored for BCD, achieving impressive results. However, most existing BCD methods focus on identifying the location of changed buildings [4], [5]. A more refined categorization

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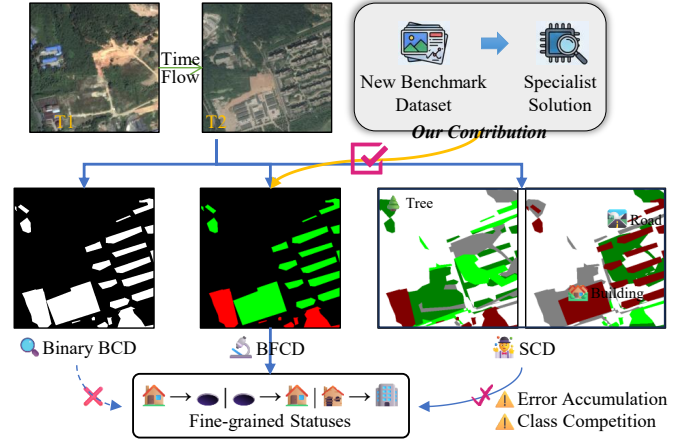


Fig. 1. The illustration of the motivation and logic of our work.

of changes, such as identifying the “demolition” or “appearance” of buildings, is often ignored. In practical applications, recognizing buildings demolished for urban expansion or those destroyed by a severe hurricane would offer valuable insights. By incorporating such fine-grained change statuses, BCD can provide more meaningful information for disaster management [6], urban planning [7], [8], etc.

Despite the potential of building fine-grained change detection (BFCD), the solutions and datasets supporting this task are still sacrificed. First, the existing datasets usually aim at specific scenes. For example, Gupta *et al.* [9] propose a large-scale building damage assessment dataset xBD, which includes four categories: “no damage,” “minor damage,” “major damage,” and “destroyed.” However, the category definitions in xBD are based on the existing buildings in the pre-temporal image, which makes it unsuitable for identifying newly built buildings in the post-temporal image. Verma *et al.* [10] develop a multi-task change detection dataset QFabric, which encompasses six land cover change types and nine change status categories. As a multi-temporal dataset, QFabric facilitates the exploration of diverse and refined change status definitions, such as buildings under construction, construction in progress, and other nuanced stages of change. Meanwhile, it also presents substantial challenges to accuracy, thereby limiting its potential applications. In contrast to previous efforts, this work focuses on bi-temporal BFCD by categorizing building changes into three semantically distinguishable statuses: disappeared, newly appeared, and replaced. Among them, the replaced class encompasses all co-located building-to-building transitions. This includes urban reconstruction, the progression

from under construction to completed, or partial structural damage caused by natural disasters or man-made conflicts. To support this research, we introduce a novel dataset, UEDC-BFCD, comprising images from three geolocations: Xi'an, China; Istanbul, Türkiye; and Ukraine. More importantly, it encompasses three types of events, urban expansion, earthquake disasters, and regional conflicts, providing a scene-rich benchmark for BFCD.

Second, although numerous change detection methods have been developed, they are not optimal for BFCD. As illustrated in Fig. 1, traditional binary BCD methods are inherently insufficient as they focus solely on identifying change locations without providing any semantic insights into the nature of the changes. Meanwhile, although semantic change detection (SCD) can categorize changes, it often treats BFCD as a complex multi-task learning problem involving simultaneous semantic segmentation and change localization. Such coupled frameworks are prone to error accumulation and class competition, where misclassifications in one sub-task propagate to the final result, leading to suboptimal performance. Besides, the simultaneous analysis of multiple categories within a unified feature space inevitably leads to class competition and semantic confusion. More importantly, BFCD aims to advance the exploration of temporal-order-aware change modeling, focusing not merely on the static semantics of land cover objects, but crucially on their dynamic change statuses. Therefore, exploring specialized solutions is necessary for BFCD. Currently, only a few studies contribute to the BFCD [11]–[13]. Tang *et al.* [12] explore the object fine-grained change detection using frequency-decoupling interaction learning. However, they only consider two types of changes, disappeared and appeared regions, while ignoring replaced regions between objects. Wang *et al.* [13] introduce a novel rebuilt change status for buildings and propose a scale- and shape-aware network to enhance the perception of such changes, but their focus is limited to urban construction scenes.

To overcome the limitations of previous change detection models, a new prompt-driven change detector (PromptCD) is developed. In detail, we adopt an architecture with two decoder branches, the same as that presented in [13]. One is to identify the changed buildings in the pre-temporal image, and the other is to identify those in the post-temporal image. Then, the building fine-grained change types can be easily obtained by combining the outputs of the two branches. Unlike other BFCD methods that treat BFCD as a multi-class classification task or as a multi-task learning problem involving building extraction and change detection [11], [12], this architecture aims to separately distinguish changed buildings from unchanged regions in both the pre- and post-temporal images, thereby simplifying BFCD into a binary classification task. However, it also raises a key issue: how to learn a discriminative representation that separately captures changed buildings within the bitemporal images? We refer to this representation as the single-temporal change embedding (STCE).

Aiming at this, PromptCD combines the absolute difference of bi-temporal features with the single-temporal feature to form the STCE. To enhance the discrimination of the STCE, inspired by the architecture DMT [14], we introduce a prompt-

guided transformer decoder (PGTD) for efficient feature refinement. It follows the design paradigm of the DETR [15] and MaskFormer [16], employing two learnable tokens to interact with the STCE. The two tokens are used as prompts to represent regions of no change and those of change. First, the pixel-wise representation of the STCE is transformed and incorporated into the prompt tokens, generating global representations for the no-change and change regions within the single-temporal image. Then, the global representations of the pre-temporal and post-temporal images are jointly injected back into their respective STCEs to enhance semantic distinction and model spatiotemporal correlations. The above two steps are called pixel-to-prompt (Pi2Pr) and prompt-to-pixel (Pr2Pi). Finally, the prompts injected with global information serve as a pixel classifier to identify changed buildings and other areas within the single-temporal image. In PGTD, the interaction between the prompts and STCE is computationally economical, thus ensuring the efficiency of the model. To summarize, the key contributions of this work are as follows:

- 1) We introduce UEDC-BFCD, a manually annotated benchmark dataset for BFCD, encompassing diverse scenarios such as urban expansion, natural disasters, and regional conflicts. It offers high-quality pixel-wise labels for three distinct building change statuses: disappeared, newly appeared, and replaced.
- 2) We propose PromptCD, a novel BFCD model incorporating the concept of STCE. STCE is introduced to separately identify changed buildings and determine their change statuses from the bitemporal images. To enhance the discriminability of STCEs and exploit spatiotemporal correlations, we further design PGTD, enabling efficient and effective refinement of STCEs.
- 3) Extensive experiments on four datasets demonstrate that PromptCD achieves superior performance. Meanwhile, we evaluate the effectiveness of PromptCD from multiple views and discuss its differences with existing change detection methods, establishing a strong benchmark for future research in BFCD.

II. RELATED WORK

A. Change Detection

Change detection is a long-established and important research topic in remote sensing and computer vision [17], [18]. Existing tasks mainly include class-specific object change detection, such as BCD, class-agnostic binary change detection, and SCD. The first two tasks can be unified as binary change detection, which focuses on identifying the location of changes. SCD not only locates the changed regions but also classifies the semantic categories in pre- and post-temporal images. Numerous methods have been developed to address these tasks [19], [20]. Among them, SCD models often utilize a multi-task learning framework to perform both semantic segmentation and change detection simultaneously, such as SCanNet [21], ChangeMask [22], etc., thus capturing semantic change types. From a semantic transformation perspective, BFCD can be regarded as a special example of SCD. Therefore, SCD models can be seamlessly employed for BFCD, as

demonstrated by MTL-CD [11]. However, their performance depends on the accuracy of change detection and semantic segmentation, which may lead to suboptimal performance due to error accumulation.

For binary change detection, existing methods aim to address issues such as bi-temporal image domain differences and sample imbalance, thereby improving change localization accuracy [23], [24]. Their effectiveness for BFCD depends on how the differences between bi-temporal images are extracted. For instance, using only the absolute difference between bi-temporal features can locate changes, but not be used to classify the statuses of the building changes. Since different categories of change statuses are inherently time-dependent, the difference representation must model the temporal order of the bitemporal images, as demonstrated by FDI-Net [12]. This is a key distinction between BFCD and other existing change detection tasks. Based on the prior work [13], we further introduce a novel change detector that essentially overcomes the limitations of most existing approaches.

B. Image Segmentation

As a pioneering work in object detection, DETR [15] introduces an end-to-end Transformer framework that employs a set of learnable object queries, eliminating the need for anchors and region proposals. Inspired by DETR, MaskFormer [16] establishes a mask classification paradigm to predict sets of masks, offering a unified framework for image segmentation tasks. Then, Mask2Former [25] introduces masked attention that controls the context range of cross-attention, effectively enhancing model performance and accelerating convergence. PEM [26] further boosts the efficiency of Mask2Former with a prototype-based cross-attention mechanism. More recently, several domain-generalized semantic segmentation models, such as Rein [27] and TQDM [28], are also developed based on the MaskFormer framework. Additionally, the mask decoder in SAM [29] uses point or box prompts as queries to interact with image embeddings and locate areas of interest, drawing on the core idea of DETR. It is precisely inspired by these works that we develop the PGTD in our model. To the best of our knowledge, PGTD is one of the few variants of DETR and MaskFormer in change detection [30], [31].

III. METHODOLOGY

A. Overview

Fig. 2 illustrates the pipeline of the proposed PromptCD, which employs two weight-shared branches to detect changed buildings in the pre- and post-temporal images, respectively. Fine-grained change statuses of buildings are then derived by combining the outputs of both branches. Overall, PromptCD can be divided into three stages: the extraction of bi-temporal features, and the construction and decoding of the STCEs.

In the first stage, a weight-shared encoder-decoder architecture is employed to extract discriminative bi-temporal features. Specifically, given a pair of bi-temporal images, I_{pre} and I_{post} , SegFormer [32] is used as the feature encoder to capture four levels of features, each with varying sizes and information. An FPN decoder [33] is then used to progressively fuse

the multi-scale features at different levels, generating four bi-temporal feature maps, $F_{i,j}$, where $i \in \{1, 2, 3, 4\}$, $j \in \{t_1, t_2\}$. Compared to an encoder-only architecture, the fusion of multi-scale features through the FPN improves the semantic discrimination of bi-temporal features, thereby enhancing the change representation in the subsequent stages.

In the second stage, the bi-temporal features are mapped to the STCEs. In the third stage, the STCEs are passed into the PGTD to predict the changed buildings in both the pre- and post-temporal images separately. Finally, the change statuses are computed using the following equation:

$$\mathcal{M}_{cs} = \mathcal{M}_{pre} + 2 * \mathcal{M}_{post}, \quad (1)$$

where $\mathcal{M}_{pre}$ and $\mathcal{M}_{post}$ are the binary masks indicating the changed buildings in the pre-temporal and post-temporal images, respectively. $\mathcal{M}_{cs}$ thus reveals the change statuses of the buildings: disappeared, newly-appeared, or replaced. Specifically, the pixel values of 0, 1, 2, and 3 in $\mathcal{M}_{cs}$ denote no change, disappeared buildings, newly-appeared buildings, and replaced buildings, respectively.

B. Single-Temporal Change Embedding

The concept of the STCE is built on the two-branch decoding architecture. It should be able to identify changed buildings within the pre- and post-temporal images separately, thereby effectively illustrating the change statuses of changed buildings. To guide the design of STCE, we revisit the definition of building change status. Taking the pre-temporal image as an example, Fig. 3 shows three examples of STCE construction. First, building change status is a time-dependent variable. From this perspective, an orderly combination of bi-temporal features can be used to construct the STCE. As shown in Fig. 3(a), a common approach for this is feature concatenation, which can be represented as:

$$\mathcal{F}_i^{pre} = \text{Concat}(F_{i,t1}, F_{i,t2}), \quad (2)$$

$$\mathcal{F}_i^{post} = \text{Concat}(F_{i,t2}, F_{i,t1}), \quad (3)$$

where F_i^{pre} and F_i^{post} can be used as the STCE of the pre-temporal and post-temporal images, respectively. F_i^{pre} is intended to locate changed buildings in the pre-temporal image, while F_i^{post} is used to detect changed buildings in the post-temporal image. Besides, as illustrated in Fig. 3(b), directed subtraction between bitemporal features is also an effective approach, which is demonstrated in the previous work [13]. Second, change status can also be defined by the specific image in which the changed buildings are located. Based on this, the STCE in PromptCD is generated by combining single-temporal features with the absolute difference between bi-temporal features. This is formulated as:

$$\mathcal{F}_i^{pre} = \text{Concat}(D_{i,t1}, F_{i,t1}), \quad (4)$$

$$\mathcal{F}_i^{post} = \text{Concat}(D_{i,t2}, F_{i,t2}). \quad (5)$$

$$D_{i,t1} = \|F_{i,t1} - F_{i,t2}\|_1, D_{i,t2} = D_{i,t1}, \quad (6)$$

where the difference representation provides the change information, while the single-temporal feature remains the spatial

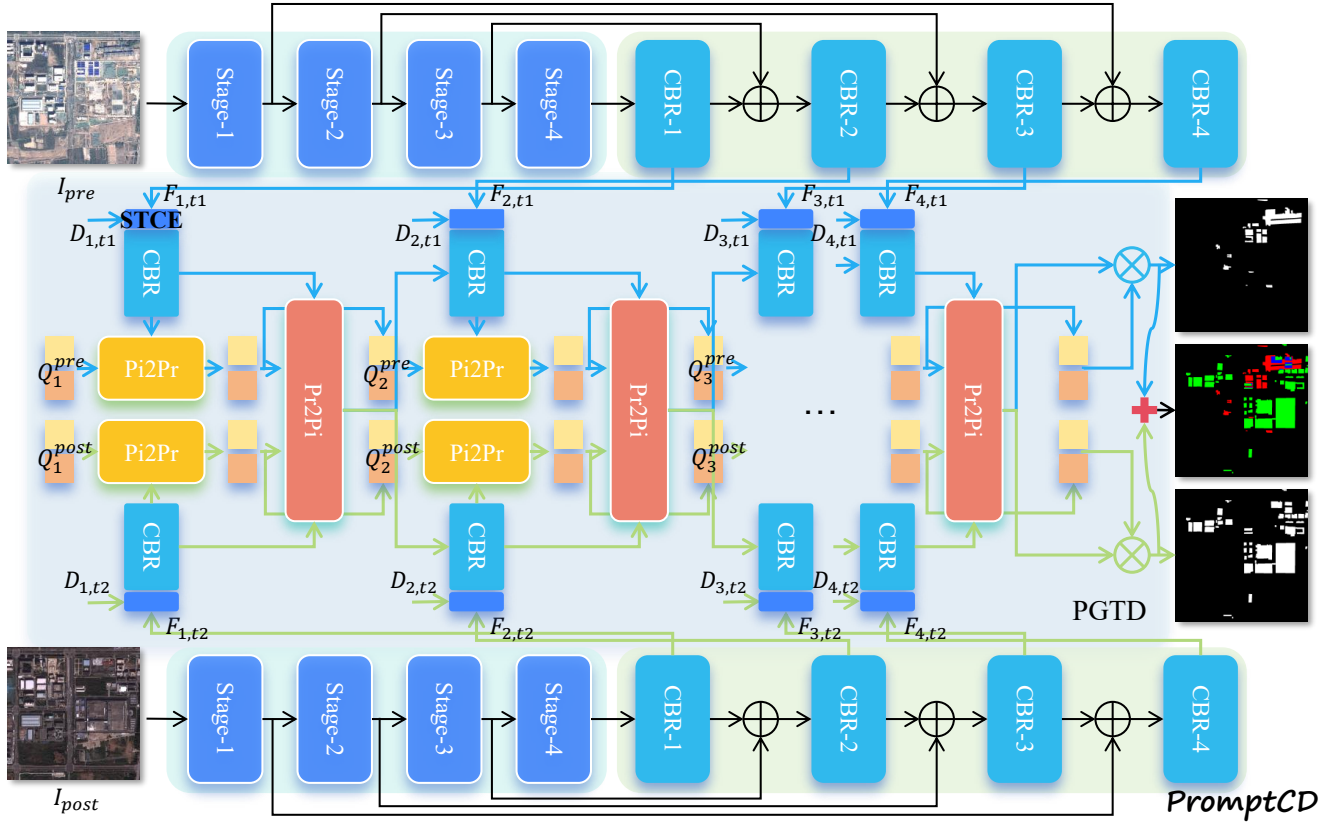


Fig. 2. Overall pipeline of the proposed PromptCD. This diagram illustrates the bi-branch architecture of PromptCD, where the blue and green arrows indicate the information flow of the bi-temporal images, respectively, during the decoding stage. The red “+” symbol represents the operation defined in Eq. 1.

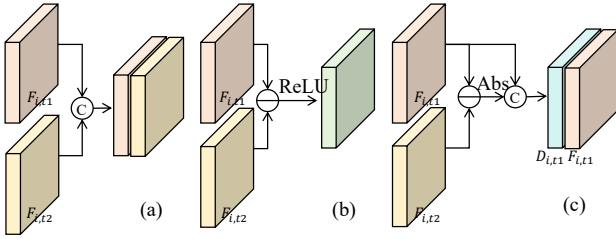


Fig. 3. Illustration of different ways to construct STCE: (a) Feature concatenation, (b) Feature subtraction with ReLU activation [13], and (c) Our.

position of the buildings. Fig. 3(c) shows our STCE construction strategy. They are fed into the PGTD to identify changed buildings in the pre-temporal and post-temporal images separately. For simplicity, we also refer to the updated representation in the PGTD as STCE.

C. Prompt-Guided Transformer Decoder

The discriminative decoding of the STCE forms the foundation for accurately locating the changed buildings and assigning their change statuses. In this context, we develop the PGTD, which leverages the same FPN structure as in the decoding of bi-temporal features, progressively fusing the multi-level STCEs to integrate multi-scale information. Diverging from conventional methods and inspired by the design paradigms of DETR and Moksformer, we introduce two learnable tokens that function as prompts for representing

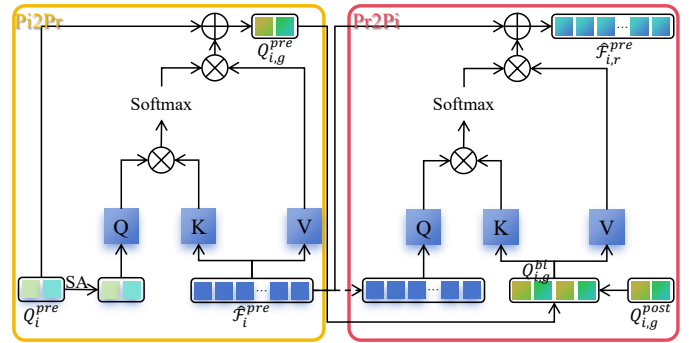


Fig. 4. The simplified schematic diagram of Pi2Pr and Pr2Pi. It presents the information flow between the prompts and the STCE.

changed buildings and other areas. These tokens interact with the STCE and are updated at each level, facilitating a dynamic refinement of the STCE and enabling the final discriminative classification. Specifically, this interaction operates in two stages: Pi2Pr and Pr2Pi. Fig. 4 presents the simplified operations of the two stages, taking the pre-temporal as an illustration. Thereafter, the pre-temporal is taken as an example to explain the detailed execution process of Pi2Pr and Pr2Pi, and the process of the post-temporal is the same.

Pi2Pr: It aims to capture the global representation of unchanged regions and changed buildings in the STCE with the guidance of the two prompts. Let $Q_i^{pre} \in \mathbb{R}^{2 \times c}$ denote the

TABLE I
COMPARISON OF RELEVANT DATASETS.

Dataset	Res.	Pairs	Task	Scenarios	Locations
GZ-CD	0.5-1m	1092	BCD	Urban Expansion	GuangZhou, China
WHU-CD	0.2m	1920	BCD	Urban Expansion	Texas, USA
LEVIR-CD	0.5m	2548	BCD	Urban Expansion	Christchurch, New Zealand
LEVIR-CD+	0.5m	3940	BCD	Urban Expansion	Texas, USA
SECOND	0.5-3m	4662	SCD	Urban Expansion	Hangzhou, Chengdu, and others, China
UE-BFCD	0.8-1m	2492	BFCD	Urban Expansion	Xi'an, China
UEDC-BFCD	0.8-1.5m	3506	BFCD	Urban Expansion, Natural Disasters, Regional Conflicts	Xi'an, China; Istanbul, Türkiye; Ukraine

prompts at the i -th level of PGTD, and let $\hat{\mathcal{F}}_i^{pre} \in \mathbb{R}^{N_i \times c}$ denote the corresponding STCE. First, a multi-head self-attention (SA) [34] layer is used to fuse the two prompts. Next, the pixel-wise feature of the STCE is incorporated into the prompts by a multi-head cross-attention (CA) layer, thus obtaining the global representation of the unchanged regions and changed buildings. The process can be formulated as follows:

$$\begin{aligned} Q_{i,g}^{pre} &= \text{Pi2Pr}_i(Q_i^{pre}, \hat{\mathcal{F}}_i^{pre}) \\ &= Q_i^{pre} + \text{MLP}(\text{CA}(\text{SA}(Q_i^{pre}), \hat{\mathcal{F}}_i^{pre})), \end{aligned} \quad (7)$$

where MLP is the multiple-layer perceptron. $Q_{i,g}^{pre}$ are the updated prompts and used at the next level, i.e., $Q_{i+1}^{pre} = Q_{i,g}^{pre}$, achieving the cross-level interaction. Notably, the prompts are the query vectors in the cross-attention layer.

Pr2Pi: This step aims to inject the global representation, i.e., the updated prompts, back into the STCE, thereby refining it and enhancing its semantic discrimination. Besides, the spatiotemporal correlation in bi-temporal STCEs is considered in Pr2Pi, which reflects the semantic complementarity of the same class objects distributed across different locations in the bi-temporal images. In detail, the prompts $Q_{i,g}^{pre}$ and $Q_{i,g}^{post}$ are first concatenated:

$$Q_{i,g}^{bi} = \text{Concat}(Q_{i,g}^{pre}, Q_{i,g}^{post}) \quad (8)$$

where $Q_{i,g}^{bi} \in \mathbb{R}^{4 \times c}$. Then they interact with the STCE through a CA layer. The formulation is denoted as:

$$\begin{aligned} \hat{\mathcal{F}}_{i,r}^{pre} &= \text{Pr2Pi}_i(\hat{\mathcal{F}}_i^{pre}, Q_{i,g}^{bi}) \\ &= \text{MLP}(\text{CA}(\hat{\mathcal{F}}_i^{pre}, Q_{i,g}^{bi})), \end{aligned} \quad (9)$$

where $\hat{\mathcal{F}}_{i,r}^{pre}$ is the refined STCE.

From the above process, the complexity of Pi2Pr is $\mathcal{O}(2N)$, while that of Pr2Pi is $\mathcal{O}(4N)$, both of which are significantly lower than the complexity $\mathcal{O}(N^2)$ using the conventional Transformer without prompt guidance. Therefore, the Pi2Pr and Pr2Pi not only effectively refine the STCEs, but also ensure the efficiency of the model.

PGTD: To summarize, the PGTD can be denoted using the following equations:

$$\hat{\mathcal{F}}_{i+1} = \text{CBR}(\text{CBR}(\mathcal{F}_{i+1}) + \hat{\mathcal{F}}_{i,r}), \quad (10)$$

$$\hat{\mathcal{F}}_1 = \text{CBR}(\text{CBR}(\mathcal{F}_1)), \quad (11)$$

$$Q_{i,g} = \text{Pi2Pr}_i(Q_i, \hat{\mathcal{F}}_i), \quad (12)$$

$$\hat{\mathcal{F}}_{i,r} = \text{Pr2Pi}_i(\hat{\mathcal{F}}_i, Q_{i,g}^{bi}), \quad (13)$$

where $i \in \{1, 2, 3, 4\}$, and CBR contains a vanilla convolution operation, a batch normalization layer, and a ReLU activation function. Finally, the logit maps $\mathcal{G} \in \mathbb{R}^{2 \times H \times W}$ of changed buildings in the pre-temporal and post-temporal images can be obtained by the inner product between the $\hat{\mathcal{F}}_{4,r}$ and $Q_{4,g}$. This is formulated as:

$$\mathcal{G}_{pre} = \text{R}(Q_{4,g}^{pre} (\hat{\mathcal{F}}_{4,r}^{pre})^T)_{\uparrow} \quad (14)$$

$$\mathcal{G}_{post} = \text{R}(Q_{4,g}^{post} (\hat{\mathcal{F}}_{4,r}^{post})^T)_{\uparrow} \quad (15)$$

where $\text{R}(\cdot)$ denotes the reshape operation. Here, it can be found that the classifier is input-adaptive and dynamic, thus helping improve the classification accuracy.

D. Loss Function

PromptCD is trained using the cross-entropy loss. Additionally, to address the class imbalance across different change statuses, the Lovász-Softmax loss [35] is incorporated. The total loss is denoted as follows:

$$\begin{aligned} \mathcal{L}_{bfcd} &= \mathcal{L}_{ce}(\mathcal{G}_{pre}, G_{pre}) + \mathcal{L}_{lov}(\mathcal{G}_{pre}, G_{pre}) \\ &\quad + \mathcal{L}_{ce}(\mathcal{G}_{post}, G_{post}) + \mathcal{L}_{lov}(\mathcal{G}_{post}, G_{post}), \end{aligned} \quad (16)$$

where G_{pre} and G_{post} represent the ground truth (GT) of the changed buildings in the pre-temporal and post-temporal images, respectively.

IV. UEDC-BFCD

This section introduces the UEDC-BFCD dataset, which comprises 3506 pairs of images with a size of 512×512 pixels and a resolution ranging from 0.8 to 1.5 meters. Table. I reports the detailed information of UEDC-BFCD and the comparison with others, where the image size is unified to 512×512 pixels. Here, we clarify that UEDC-BFCD is not a large-scale dataset. However, compared to existing representative datasets, it is sufficiently comprehensive to support deep learning-based model development. First, this dataset covers urban expansion, natural disasters, and regional conflicts three scenarios, providing a broader domain diversity compared to many widely used benchmark datasets in the BCD task. Second, the annotation of the building change statuses is fully manual. We form a team, with each member responsible for annotating different parts of the data. After completing the annotations, the annotated data is exchanged sequentially for

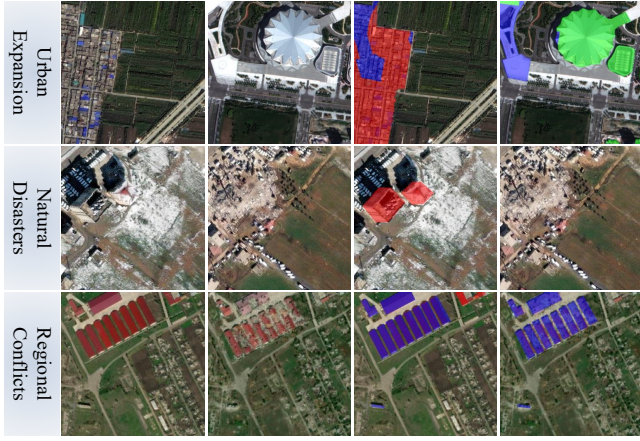


Fig. 5. Representative examples from our UEDC-BFCD dataset. The labels are overlaid on the original images to clearly indicate the status of different changed buildings. The blue, red, and green regions indicate buildings that have disappeared, newly appeared, and been replaced, respectively.

review and correction, helping to improve the reliability of the annotation quality. Fig. 5 illustrates three representative examples, including demolished villages, buildings collapsed into ruins, and factories damaged by conflict.

The urban expansion scenario includes 2,492 image pairs, acquired using the GaoFen-2 (GF2) and SuperView-2 (SV2) satellites, focusing on the Xi'an region in China. The images cover four years: 2016, 2017, 2018, and 2022, and have undergone necessary spatial alignment. Importantly, the alignment process is fully automated. Although no manual intervention exists in the alignment process, the results remain acceptable. The slight registration discrepancies in some areas increase the challenge of this dataset, while also encouraging researchers to consider the misalignment factor between the bi-temporal images when developing change detection methods. This subset is also released as a standalone dataset named UE-BFCD, specifically targeting the urban expansion domain.

The natural disaster scenario consists of 708 image pairs, originally introduced in our prior work [36]. These image pairs depict the pre- and post-earthquake conditions in Türkiye during 2023, acquired from Maxar's open data program. This earthquake event resulted in the collapse of numerous buildings. Furthermore, the regional conflict scenario contains 306 image pairs, collected from the ESRI platform, which supports academic use of high-resolution satellite imagery. These images focus on the Russia-Ukraine conflict, offering a real-world dataset for analyzing infrastructure damage in conflict zones. In particular, these three scenarios in UEDC-BFCD can support a wide range of applications, including urban planning and damage assessment, among others.

V. EXPERIMENTS

A. Datasets and Evaluation Metrics

Datasets: We harness four datasets to evaluate the performance of state-of-the-art change detection methods and the proposed PromptCD. Two of them are our UEDC-BFCD and its subset UE-BFCD, which focuses specifically on urban-evolution scenarios. They are divided into training and testing

sets: UE-BFCD contains 1,772 image pairs for training and 720 pairs for testing, while UEDC-BFCD consists of 2,388 pairs for training and 1,118 pairs for testing. The other two datasets are SECOND [37] and WHU-CD. SECOND is a SCD dataset, which includes six types of land cover. For BFCD, it is modified by retaining only the regions corresponding to changed buildings in the bi-temporal images. Each image has a resolution of 512×512 pixels. Following the official data split, 2,920 image pairs are used for training and 1,681 pairs for testing. WHU-CD, originally a BCD dataset, has been relabeled for BFCD in prior work [13]. All images are cropped to a size of 512×512 pixels. Following the official split [38], 1,260 image pairs are used for training, with the remainder reserved for testing.

Evaluation Metrics: F1-score (F1) and the intersection of union (IoU) are used to evaluate the accuracy of a single change status prediction, while mF1 and mIoU are used to assess the overall accuracy of the model. The definitions of the F1-score and IoU metrics can be found in [39]. Furthermore, since BFCD depends on temporal order, buildings originally classified as 'disappeared' or 'newly-appeared' should be reclassified as 'newly-appeared' or 'disappeared' after swapping the temporal order of bi-temporal images. To assess the robustness of current methods in detecting change statuses after temporal swapping, their $mF1_{t_{21}}$ and $mIoU_{t_{21}}$ are compared, where t_{21} denotes the reversed temporal order. The complexity of the model is assessed based on its number of parameters (Params) and floating-point operations (FLOPs). FLOPs are measured at resolution 256×256 .

B. Implementation Details

Our model is trained using the AdamW optimizer, with β_1 set to 0.9 and β_2 set to 0.99. The initial learning rate is 0.0001, adjusted using a linear decay strategy. The batch size is set to 4, and the total number of training epochs is 100. For the extraction of bi-temporal features, we use the SegFormer MiT-b0 as the encoder due to its efficiency in handling multi-scale features. Additionally, the number of attention heads in Pi2Pr and Pr2Pi is set to 4, while the feature dimension in both the bi-temporal feature decoder and the PGTD is unified to 128. All experiments are conducted on an RTX 3090 GPU.

C. Ablation Study

This section thoroughly investigates the effectiveness of the proposed PromptCD. Comprehensive ablation studies are conducted on the UE-BFCD dataset to investigate the effectiveness of each core component in PromptCD and the impact of its structural variants.

Components of PromptCD: We present a step-by-step investigation of the construction of PromptCD to highlight the effectiveness of its components. Specifically, we analyze the bi-temporal feature decoder (Bi-D), the STCE decoder (STCE-D), and the loss function. The experimental results are shown in Table II. The baseline model uses a linear transformation (LT) as the Bi-D and an MLP as the STCE-D. Subsequently, LT and MLP are sequentially replaced with FPN. From the first four rows, we observe a progressive

TABLE II

THE ABLATION STUDIES OF PROMPTCD ON THE UE-BFCD DATASET. THE BEST-PERFORMING RESULTS ARE HIGHLIGHTED IN BOLD (%).

Bi-D	Method		Disappeared		Newly-Appeared		Replaced		No Change		Overall		Complexity	
	STCE-D	Loss	F1	IoU	F1	IoU	F1	IoU	F1	IoU	mF1	mIoU	Params	FLOPs
LT	MLP	$\mathcal{L}_{ce}$	78.33	64.39	59.82	42.67	48.35	31.89	98.84	97.70	71.34	59.16	4.63M	10.4G
LT	FPN	$\mathcal{L}_{ce}$	78.60	64.74	59.30	42.11	49.19	32.61	98.85	97.77	71.48	59.31	4.22M	2.24G
FPN	MLP	$\mathcal{L}_{ce}$	79.57	66.07	59.99	42.85	51.39	34.58	98.88	97.80	72.46	60.32	5.34M	12.1G
FPN	FPN	$\mathcal{L}_{ce}$	79.52	66.00	62.06	44.99	50.48	33.76	98.91	97.85	72.74	60.65	4.93M	3.87G
FPN	FPN	$\mathcal{L}_{ce}+\mathcal{L}_{lov}$	79.65	66.18	63.28	46.28	52.10	35.23	98.87	97.76	73.47	61.36	4.93M	3.87G
FPN	FPN+	$\mathcal{L}_{ce}+\mathcal{L}_{lov}$	79.52	66.01	63.89	46.94	51.93	35.07	98.89	97.80	73.56	61.45	6.11M	7.08G
FPN	PGTD	$\mathcal{L}_{ce}+\mathcal{L}_{lov}$	79.50	65.98	64.01	47.07	54.47	37.43	98.86	97.77	74.21	62.05	5.99M	4.23G

TABLE III

THE EFFECT OF INSERTING Pi2Pr AND Pr2Pi AT DIFFERENT LEVELS IN PGTD. THE BEST-PERFORMING RESULTS ARE HIGHLIGHTED IN BOLD (%).

PGTD				Evaluation Metrics			
L4	L3	L2	L1	mF1	mIoU	Params	FLOPs
✓				73.53	61.44	5.19M	4.14G
✓	✓			73.81	61.69	5.46M	4.20G
✓	✓	✓		73.91	61.78	5.73M	4.22G
✓	✓	✓	✓	74.21	62.05	5.99M	4.23G

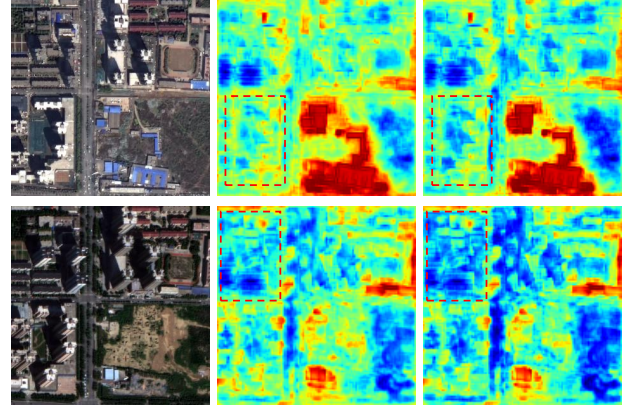
TABLE IV

THE ROLE OF COMPONENTS IN PGTD. THE BEST-PERFORMING RESULTS ARE HIGHLIGHTED IN BOLD (%).

Method	mF1	mIoU	Params	FLOPs
PGTD w/o Pr2Pi _{1,2,3}	73.51	61.39	5.69M	4.14G
Pi2Pr w/o SA	73.40	61.31	5.73M	4.23G
Pr2Pi w/ SA	73.60	61.50	6.26M	4.23G
Pr2Pi w/o Eq. 8	73.82	61.65	5.99M	4.23G
STCE w/ Eqs. 2 & 3	73.46	61.43	5.99M	4.23G
PromptCD	74.21	62.05	5.99M	4.23G

improvement in the overall performance. In particular, FLOPs are significantly reduced after replacing the MLP STCE-D with FPN. Meanwhile, after introducing the FPN as the Bi-D, both mF1 and mIoU improve by approximately 1.1%. It is confirmed that effective multi-scale feature fusion during the bi-temporal feature extraction stage is helpful. Thereafter, the loss function integrates the Lovász-Softmax loss to alleviate the imbalance between the no-change category and other statuses. The performance of the model further improves. Finally, the proposed PGTD is introduced, which improves F1 by 1.73% on the "appeared" status and 2.37% on the "rebuilt" status. The accuracy of the other two statuses remains comparable. To isolate the impact of model capacity, a scale-aligned variant denoted as FPN+ is implemented by adding linear layers to match the computational scale of PGTD. Table. II shows that FPN+ yields negligible improvements. This confirms that the advantages of PGTD arise from the architectural design rather than increased model complexity. To conclude, the disassembly of PromptCD highlights the necessity of its components.

Variants of PGTD: We first evaluate the multi-level configuration of Pi2Pr and Pr2Pi. (As reported in Table III, applying both components across all four levels yields the best performance, outperforming the last-level-only baseline



Pre & Post Image before Pi2Pr & Pr2Pi after Pi2Pr & Pr2Pi

Fig. 6. Visualizing the qualitative effectiveness of Pi2Pr and Pr2Pi. The regions bounded by the red boxes clearly show that the STCE is effectively refined through the Pi2Pr and Pr2Pi operations.

by 0.68% in mF1 with a negligible FLOPs increase of 0.09G. This gain stems from cross-level interactions facilitated by layer-wise prompt updates in Pi2Pr. Next, we explore the effectiveness of Pr2Pi and the introduced STCE. The results are reported in the Table. IV. Removing the Pr2Pi components from the first three levels causes a performance decline, which demonstrates that Pr2Pi can indeed refine the STCEs. Pr2Pi₄ is retained to ensure model convergence when prompts are used as the classifier, as our additional experiments show that the model fails to converge without it. Crucially, the empirical findings confirm that employing self-attention is effective for capturing dependencies in the Pi2Pr branch, while omitting it in the Pr2Pi branch avoids performance degradation. Additionally, removing the joint between the pre- and post-temporal prompts in Pr2Pi also reduces accuracy, highlighting the importance of spatiotemporal correlation modeling. Finally, the effectiveness of using bi-temporal feature concatenation to generate the STCE is investigated. Replacing our design with standard concatenation decreases the mF1 by 0.75%. Qualitative evidence visualized in Fig. 6 further confirms that Pi2Pr and Pr2Pi effectively enhance the discrimination of changed buildings while suppressing unchanged backgrounds, verifying the overall efficacy of PGTD.

D. Comparison Results

In this section, we compare the proposed PromptCD with nine state-of-the-art change detection methods to establish

TABLE V
QUANTITATIVE COMPARISON OF METHODS ON THE FOUR DATASETS, WITH THE BEST RESULTS IN BOLD (%).

Method	Backbone	UE-BFCD			UEDC-BFCD			SECOND			WHU-CD			Complexity	
		mF1	mIoU	mIoU _c	mF1	mIoU	mIoU _c	mF1	mIoU	mIoU _c	mF1	mIoU	mIoU _c	Params	FLOPs
ChangeFormer [40]	MiT-b0	72.32	60.18	47.64	69.10	57.27	43.71	67.68	56.90	44.36	81.02	71.25	61.85	3.85M	2.46G
CSINet [41]	ResNet-18	70.10	57.98	44.75	66.67	54.66	40.26	66.02	54.95	41.95	78.40	68.29	57.95	57.6M	367G
BAN-BiT [42]	ResNet18+ViT-b	71.12	58.93	46.03	68.42	56.39	39.89	66.37	56.61	42.80	77.75	68.49	58.19	68.2M	36.3G
BAN-CF [42]	MiT-b0+ViT-b	71.71	59.58	46.86	68.89	56.84	38.60	66.70	56.26	43.50	79.20	70.06	60.26	68.8M	29.2G
CASP [43]	MiT-b0	72.55	60.43	47.96	70.58	58.44	45.25	67.64	57.39	44.91	79.22	70.11	60.35	4.90M	3.99G
MPCD [44]	Mamba	71.83	59.90	47.26	71.11	59.01	46.04	67.87	57.10	45.29	82.29	72.23	60.10	41.93M	52.87G
MSCANet [45]	ResNet18	68.11	55.95	42.11	62.48	51.06	35.60	63.86	53.19	39.70	66.61	58.38	44.89	16.59M	14.91G
FDINet [12]	EfficientNet-b4	72.07	59.91	47.29	69.78	57.81	44.43	67.87	57.10	44.62	82.29	72.23	63.21	2.36M	4.07G
S ² Net [13]	MiT-b0	71.78	59.69	47.00	68.58	56.66	42.90	67.30	56.83	44.24	86.70	77.90	70.71	4.95M	5.96G
PromptCD	MiT-b0	74.21	62.05	50.16	70.87	58.81	45.75	70.44	59.38	47.51	87.70	79.30	72.58	5.99M	4.23G

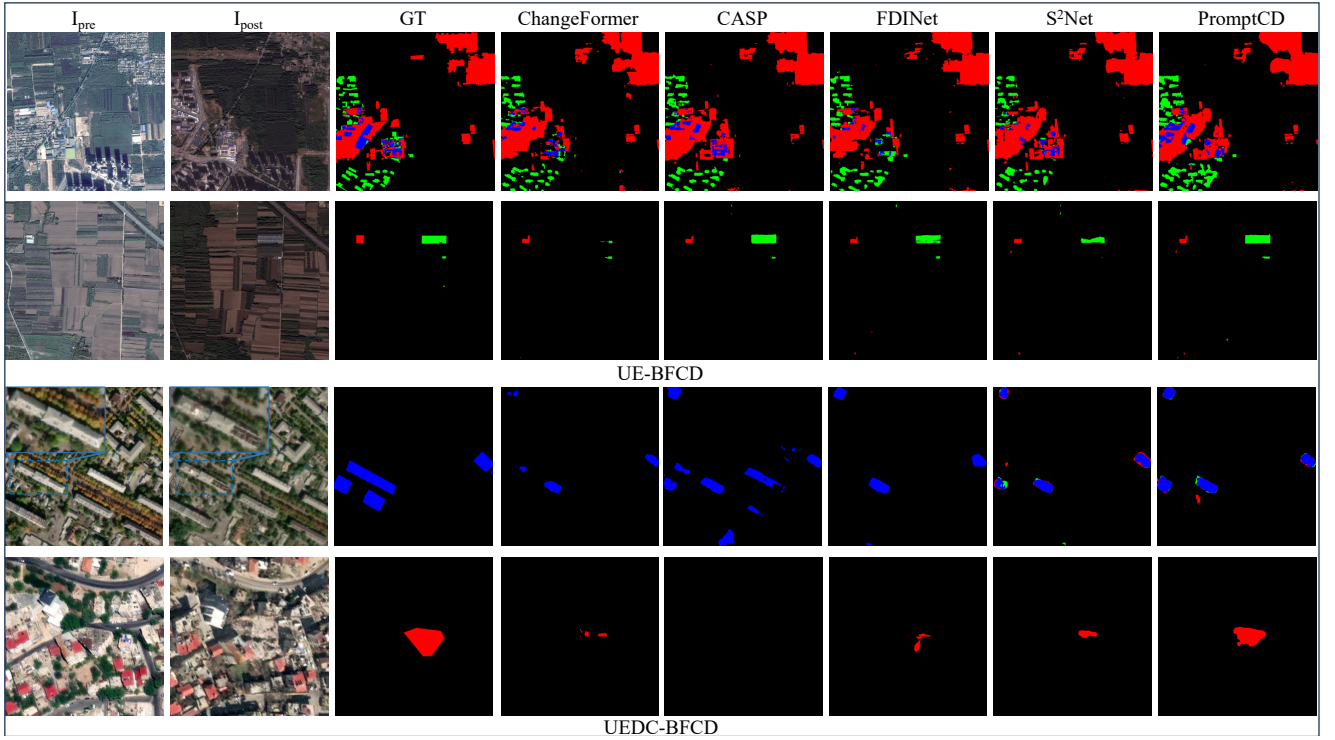


Fig. 7. The visual comparison of several representative methods on the UE-BFCD and UEDC-BFCD datasets. Note that the red, green, and blue colors denote the disappeared, newly appeared, and replaced building regions, respectively.

an evaluation benchmark for BFCD. The compared methods include ChangeFormer [40], CSINet [41], BAN-BiT [42], BAN-CF [42], CASP [43], MPCD [44], MSCANet [45], FDINet [12] and S²Net [13]. Among them, FDINet and S²Net are dedicated models for fine-grained change detection. The other methods are designed for binary change detection, and their applicability to BFCD is examined here. To ensure a fair comparison, the backbone networks in these methods are kept lightweight according to their respective configurations.

Quantitative Results: The experimental results are presented in Table V. The compared methods demonstrate distinct performance characteristics for the BFCD task. Although several models, such as the fine-grained change detection network MSCANet, are not natively designed for BCD, they can be adapted to this task to serve as reference benchmarks. Across the evaluated datasets, the proposed PromptCD delivers competitive performance compared to these networks in terms

of mF1, mIoU, and the change-category mIoU_c. Specifically, on the UE-BFCD dataset, PromptCD achieves an mF1 of 74.21% and an mIoU of 62.05%, outperforming CASP by 1.66% and 1.62%, respectively. On the multi-scenario UEDC-BFCD dataset, the performance of all methods experiences a decline due to the increased intra-class variations. In this context, the Mamba-based model, MPCD, achieves the highest indicators with 41.93M parameters and 52.87G FLOPs. Conversely, PromptCD yields comparable results, obtaining an mF1 of 70.87% and a mIoU_c of 45.75%, while utilizing only 5.99M parameters and 4.23G FLOPs. Meanwhile, compared with MSCANet, PromptCD maintains a higher accuracy across the evaluation metrics, showing the benefit of the proposed architecture tailored for BFCD. Furthermore, the evaluation on the WHU-CD and SECOND datasets shows the generalization capability of the proposed method. On the WHU-CD dataset, PromptCD surpasses S²Net by 1.00% in mF1 and

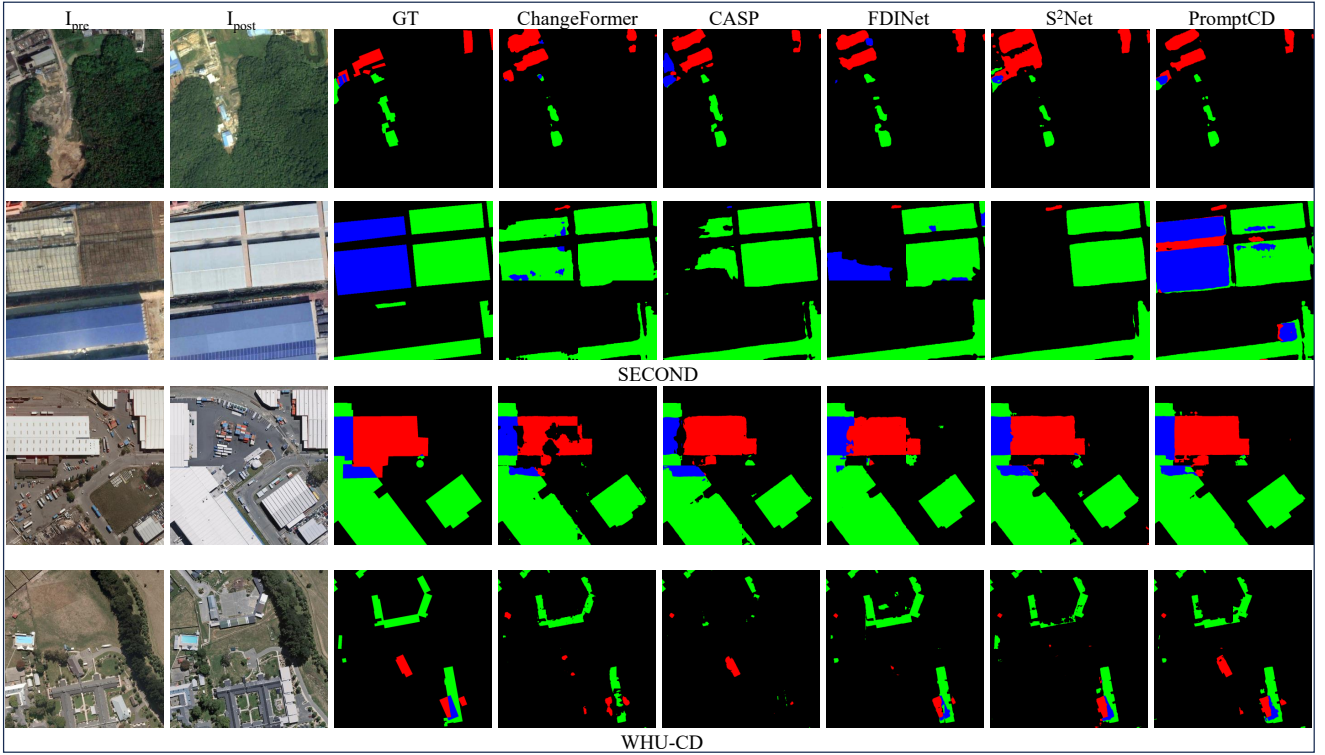


Fig. 8. The visual comparison of several representative methods on the SECOND and WHU-CD datasets. Note that the red, green, and blue colors denote the disappeared, newly appeared, and replaced building regions, respectively.

1.87% in the $mIoU_c$ score. This consistent performance across distinct benchmarks indicates that the architectural design of PromptCD is effective under different style variations and spatial distributions. To further validate the performance, the proposed PromptCD is compared with several representative SCD models on the SECOND dataset, with additional quantitative results provided in the supplementary material.

Visualized Results: Figs. 7 and 8 present qualitative comparison results, showcasing eight representative examples from different datasets that highlight diverse types of changes. As shown in Fig. 7, the UE-BFCD examples depict demolished villages and newly constructed greenhouses. The two image pairs from UEDC-BFCD illustrate urban residential areas damaged by man-made conflicts and those with collapsed buildings caused by the earthquake. In the first pair from UE-BFCD, most methods fail to detect changes in replaced regions where buildings of different structures occupy the same location in the pre- and post-temporal images. Meanwhile, on UEDC-BFCD, PromptCD accurately detects locally damaged buildings in conflict zones, with fewer omissions and false alarms. In the second UEDC-BFCD example, PromptCD is the only method that identifies the collapsed buildings with relatively high completeness. Such accuracy is essential for effective response and relief operations in disaster-affected areas. Fig. 8 shows the four visual results of the SECOND and WHU-CD datasets. In the second example on the SECOND dataset, two buildings are under construction in the pre-temporal image and fully built in the post-temporal image. They are considered replaced regions. Only PromptCD correctly detects their status. Although there is a little false

detection in the visualized examples, PromptCD presents the best results overall. Meanwhile, for the given examples from the WHU-CD dataset, our method still shows competitive performance. Overall, the above comparisons from quantitative and qualitative views fully demonstrate the effectiveness of PromptCD under diverse scenarios.

VI. CONCLUSION

In this work, we explore BFCD by modeling change statuses instead of just localization. First, we introduce the UEDC-BFCD dataset for BFCD across three types of events, urban expansion, natural disasters, and regional conflicts. In addition, we propose PromptCD, a model that accurately and efficiently identifies changed buildings in a single-temporal image through its key components, STCE and PGTD. Then, change statuses are derived by combining changed buildings in pre- and post-temporal images. Extensive experiments show the superiority of PromptCD. More importantly, experiments reveal that while existing binary change detection and SCD methods can be used for BFCD, they are suboptimal, underscoring the importance of this work. A limitation of this study is that the change statuses are not sufficiently refined. In the future, we will expand the bi-temporal BFCD benchmark with more challenging change statuses and diverse scenes.

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Supplementary Material

TABLE I
QUANTITATIVE COMPARISON WITH ADVANCED SCD METHODS ON THE
SECOND DATASETS. THE BEST-PERFORMING RESULTS ARE
HIGHLIGHTED IN BOLD (%).

Method	mF1	mIoU	mF1 _{t₂₁}	mIoU _{t₂₁}	Params	FLOPs
SSCDI [1]	68.23	57.10	67.13	56.06	23.31M	47.44G
BiSRNet [1]	67.48	56.54	66.64	56.55	23.38M	47.54G
TED [2]	68.33	57.28	67.43	56.37	24.19M	51.07G
SCanNet [2]	66.24	55.99	66.30	55.49	27.90M	66.24G
HGINet [3]	65.77	55.47	64.88	54.75	27.70M	12.65G
BTSCD [4]	69.70	58.66	69.69	58.46	24.91M	57.06G
PromptCD	70.44	59.38	70.44	59.38	5.99M	4.23G

A. Comparison with SCD methods

We investigate the performance of several advanced SCD methods for BFCd, including SSCDI [1], BiSRNet [1], TED [2], SCanNet [2], HGINet [3], and BTSCD [4]. Table. I presents the comparison results on the SECOND dataset. We directly employ the models trained for SCD to validate their performance for BFCd. It can be found that all compared methods are inferior to PromptCD. First, their computational complexity is significantly higher than ours, primarily due to the multi-task framework. Second, they cannot maintain consistent performance before and after temporal swapping, increasing the risk of degradation in diverse scenarios. Third, the multi-task learning framework also leads to error accumulation, ultimately resulting in suboptimal results. Although PromptCD employs a two-branch decoding architecture, it essentially performs single-task learning. The introduction of PromptCD helps push the performance boundary for BFCd, and its design strategy, specifically tailored for BFCd, merits further exploration.

B. Temporal Swap Sensitivity

This part aims to explore the robustness of the model when the temporal order of bitemporal images is swapped. In detail, the original pre-temporal image I_{pre} is regarded as the post-temporal image " I_{post} ", while the original post-temporal image I_{post} is treated as the pre-temporal image " I_{pre} ". The actual change statuses of the buildings also adjust accordingly. We focus on analyzing previous binary change detection methods, and the quantitative results on the UEDC-BFCd dataset are reported in Table II.

We first evaluate the model trained following the original temporal order without temporal exchange augmentation (w/o EX). After swapping the temporal order, the performance of all compared binary change detection methods, including FDI Net, declines. In particular, although FDI Net introduces the temporal exchange consistency learning technique, it fails to adapt to temporally reversed inputs. Furthermore, MPCD similarly exhibits a severe performance degradation

upon temporal reversal, despite its optimal accuracy when restricted to the original forward transition, i.e., from training-time pre-temporal to post-temporal images. This asymmetric behavior indirectly demonstrates its failure to learn true temporal-order-dependent fine-grained representations. This performance degradation likely stems from the inherent design of their difference extraction mechanisms. Specifically, due to the distinct domain characteristics of bi-temporal images, swapping the temporal order induces a domain shift within the resulting difference representations. Consequently, these methods fail to generalize to such representational variations, leading to a decline in accuracy.

How can the robustness of existing methods to temporal swapping be improved? We introduce a simple training strategy, i.e., temporal exchange augmentation (w/ EX). As shown in Table II, this approach effectively enhances model robustness against temporally reversed inputs. However, most models experience a performance trade-off, compromising accuracy on the original temporal order. In contrast, PromptCD and S²Net avoid this issue owing to their two-branch decoder architectures. Moreover, PromptCD consistently outperforms other methods, further demonstrating its competitiveness.

C. Cross Scenario and Cross-resolution Generalization

To evaluate the transferability of our model, we conduct a generalization test by deploying a variant trained exclusively on the standard Urban Expansion dataset directly to the unseen Regional Conflict scenario, as compiled in Table III. The results reveal a distinct performance bifurcation. On one hand, the model exhibits a catastrophic drop in fine-grained change categories, with its accuracy on the Replaced class plunging to 0.00%. This degradation underscores a key limitation, proving that subtle structural features like conflict-specific ruins are highly scenario-dependent and resistant to direct zero-shot transfer. On the other hand, the model maintains a highly comparable performance to the comprehensive UEDC-trained baseline in both the Disappeared and Newly-Appeared categories, yielding 12.63% versus 12.57% and 1.77% versus 2.14% respectively. This persistence implies that macro object-level alterations, such as complete demolition or new construction, share generic visual primitives across domains, granting the framework an intrinsic geometric robustness.

Concurrently, we investigate the sensitivity of the model to input scale variations by evaluating its performance under different inference resolutions in Table. IV. The empirical results unveil a pronounced dependency on high-resolution data: compressing the input from 512×512 to 256×256 triggers a severe performance drop, with the mF1 and mIoU falling from 74.21% and 62.05% to 61.74% and 50.28%, respectively. This outcome delineates another critical boundary of our framework. Because building functional change detection

TABLE II
QUANTITATIVE COMPARISON OF THE TEMPORAL SWAPPING SENSITIVITY, WITH THE BEST RESULTS IN BOLD (%).

Method	w/o EX				w/ EX			
	mF1	mIoU	mF1 _{t₂₁}	mIoU _{t₂₁}	mF1	mIoU	mF1 _{t₂₁}	mIoU _{t₂₁}
ChangeFormer [5]	69.10	57.27	47.16	38.55	68.34	56.64	68.22	56.48
CSINet [6]	66.67	54.66	43.51	35.71	65.80	53.86	68.70	56.49
BAN-BiT [7]	68.42	56.39	48.13	39.49	67.93	55.83	67.94	55.85
BAN-CF [7]	68.89	56.84	49.08	40.35	69.14	57.11	69.05	57.09
CASP [8]	70.58	58.44	44.73	36.56	69.79	57.78	70.04	58.00
MPCD [9]	71.11	59.01	50.39	41.35	70.50	58.15	49.89	41.43
FDINet [10]	69.78	57.81	50.05	41.73	69.88	57.79	50.90	42.39
S ² Net [11]	68.58	56.66	68.58	56.66	68.58	56.66	68.58	56.66
PromptCD	70.87	58.81	70.87	58.81	70.87	58.81	70.87	58.81

TABLE III
QUANTITATIVE RESULTS ON CROSS-SCENARIOS GENERALIZATION (%).

Scenario	Dataset	Disappeared	Newly-Appeared	Replaced	Unchanged
Conflict	UEDC	12.57	2.14	29.57	98.08
	UE	12.63	1.77	0.00	98.07

TABLE IV
EVALUATION RESULTS UNDER DIFFERENT INFERENCE RESOLUTIONS (%).

Inference Resolution	mF1	mIoU
512 × 512	74.21	62.05
256 × 256	61.74	50.28

inherently relies on sharp geometric outlines and localized spatial texturing, resolution downscaling severely blurs these crucial high-frequency structural clues. Consequently, the capacity of the model to resolve subtle functional attributes becomes constrained, reflecting a clear sensitivity to resolution variations during practical deployments.

D. Extension to Long-term Images

To demonstrate the scalability of our framework to multi-temporal scenarios, we seamlessly extend the proposed bi-temporal model to sequential building growth detection with minimal structural overhead. Specifically, given a sequence of T multi-temporal images $\mathcal{I} = \{I_0, I_1, \dots, I_{T-1}\}$, all frames are initially processed by a shared weight-tied backbone to extract multi-level feature representations. For each adjacent temporal pair (I_t, I_{t+1}) , we construct the STCE by concatenating the absolute feature discrepancy and the post-temporal feature map, formulated as:

$$\mathcal{F}^{t+1} = \text{Concat}(|F_{t+1} - F_t|, F_{t+1}), \quad (1)$$

where $\text{Concat}(\cdot)$ denotes the channel-wise concatenation operator. This representation is subsequently appended to the same change-aware decoder inherited from the original bi-temporal architecture, yielding $T - 1$ timestamp-change predictions. Each intermediate prediction explicitly quantifies whether a building newly emerges within the corresponding interval. The final multi-stage change map is derived by aggregating these sequential outputs, where label 0 denotes the absence of

structural changes, and labels $1, \dots, T - 1$ denote the specific construction timestamps.

We further provide qualitative results on the SpaceNet7 dataset to validate this extension (see Fig. 1). Each sample contains nine monthly images from 2018 to 2020. The visualization compares the input temporal sequence, ground-truth multi-stage labels (GT), our predictions (Pred), and their overlays on the reference image. As shown, the predicted change phases align reasonably well with the ground truth, correctly capturing the progressive construction of buildings across different time intervals. These results confirm that our bi-temporal framework can be directly generalized to multi-temporal, fine-grained change analysis, and can serve as a feasible baseline for future research on continuous time-series change detection.

In fact, multi-temporal BFCD encompasses diverse scenarios, where sequential growth detection represents only a specific instance. In more general multi-temporal applications, the fine-grained status of a changing building over a specific interval exhibits a pronounced temporal dependence, as it is intricately linked to both preceding and succeeding contexts. Despite these inherent complexities, our framework remains conceptually viable for scaling to phase-wise, fine-grained classification and identification from a technical standpoint, largely owing to the modular flexibility of our core decoupling strategy. However, a major bottleneck currently impeding the empirical validation is the scarcity of high-quality public benchmarks and annotated datasets tailored for multi-temporal, fine-grained building change statuses within the remote sensing community. Addressing this data scarcity and exploring advanced temporal modeling paradigms present a promising avenue for future research.

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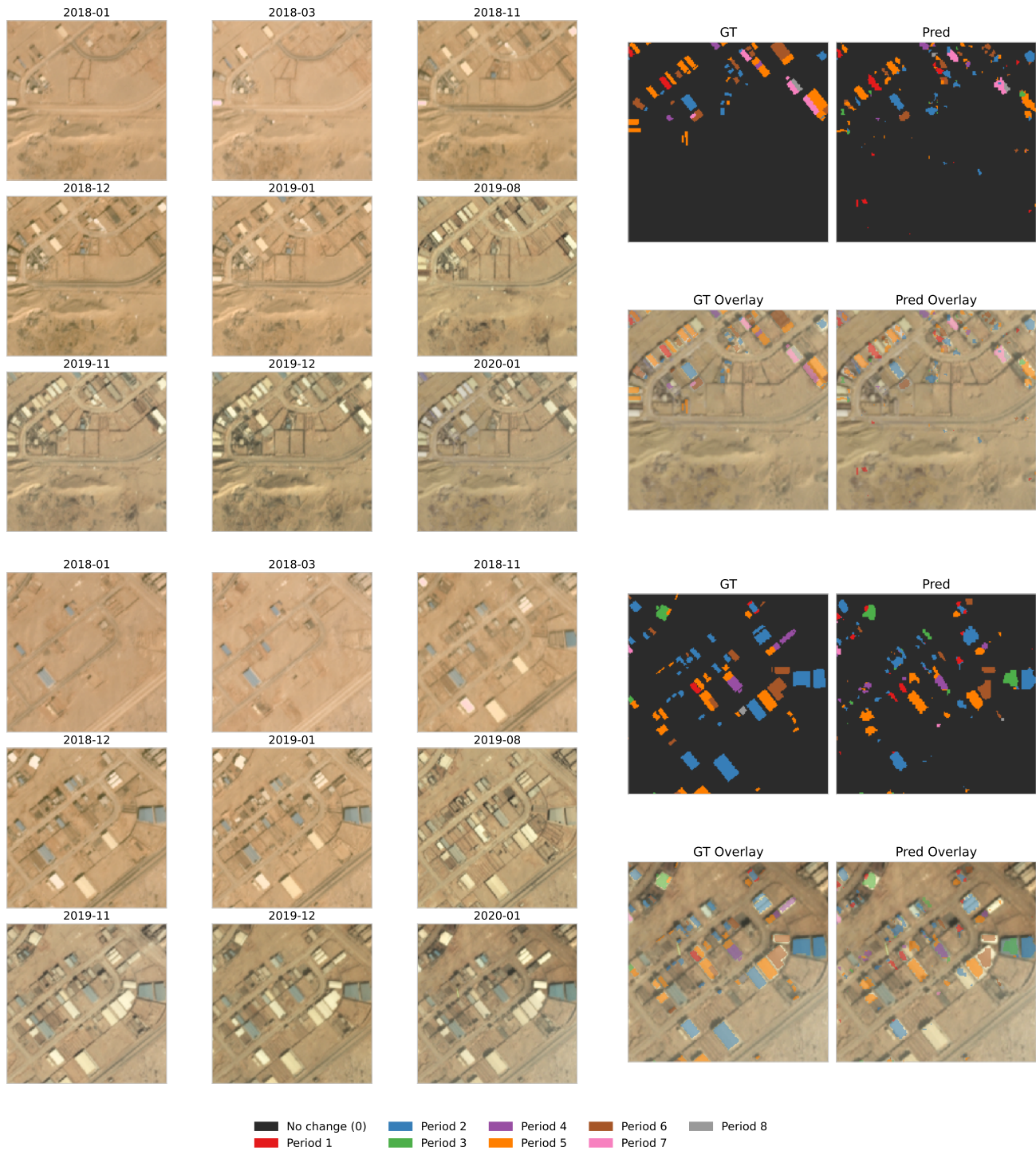


Fig. 1. Visual results on the SpaceNet7 dataset. "Period 1" denotes newly appeared buildings in the second timestamp of the nine monthly imagery sequences, with subsequent periods (e.g., Period 2, 3) following the same chronological order.

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